# ABC COLLEGE OF ENGINEERING

Kozhikode, Kerala – 673601

## EXAMINATION ANSWER PAPER

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| **Department** | : | **ARTIFICIAL INTELLIGENCE AND DATA SCIENCE** |
| **Subject** | : | **EIA** |
| **Course Code** | : | **AIT401** |
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| **Date** | : | |

2] The environmental impact assessment the public participation plays a major role because while designing a project, it can cause harm to human beings in many ways. People can lose their land, property, habitat etc. It can even cause harm to their health issues due to air pollution, water pollution, noise pollution etc. It can cause harm to the biodiversity. The public participation is important because the publics can express their opinion and tell their concerns. Then, the project can be done by meeting the needs of the people.

3] Noise pollution is a type of pollution that occurs thro noise. That is it can be from the noise of horns of vehicles, machinery noises, noise of vehicles etc. These noises can effect the ears of living beings which can lead to becoming deaf. It can also effect the ecosystem by disturbing the environment using excessiv noises. Using horns while necessary reduces the impact of noise pollution. Also reduce the use of machines that

produce large noises.

4] i) Impact prediction refers to the process of identifying the severity or extend of the impact. After identifying the the impact as positive or negative we perform impact prediction which helps to understand the severity and take precautions according to it.

ii) Socio economic impact refers to the impact that causes harm to the social and economic part which includes society, ~~human beings~~ peoples, economic etc. This can be reduced by taking safety measures like rehabilation, ~~or~~ following the rules and laws of the environment.

5] The two impacts on the water environment due to any solid waste management project include :-

* Underground water - ~~Causes~~ When the waste materials degrade to the ground, the chemicals ~~in the ground~~ contained in it gets dissolved with underground water which can cause health issues and other harms.

mitigation measure - Use proper waste management plans and avoid putting waste in the ground directly.

* Surface water - When the waste materials are thrown to the water bodies like river, ponds, lakes etc. the chemicals in the waste get dissolved in the water which also leads to many health issues as well as other harms.

mitigation measure - Avoid throwing waste to the side of river bodies like river, ponds, lakes and make use of waste bins and other waste management plans.

6] Impacts caused due to quarry mining project include :

* Noise pollution - Blasting, drilling etc can make large noises causing harm to people as well as workers. Heavy noise of machinery can also cause harm.

* Air pollution - While drilling, blasting the air gets polluted by dust particles and cause harm to people nearby as well as workers.

* Water pollution - The underground water can be affected by doing quarry mining process.

* Loss of plantation - Plants and trees are removed from the places which can cause huge amount of damage to the ecosystem.

11] Lets consider the current highway widening project in Kerala which NH66.

* Objectives :-
  - Traffic reduction
  - Easy transportation
  - Less noise pollution
  - Less wasting of time
  - Reduce air pollution

* Baseline study
Collects all the details about the current status of air, water, noise and other data to study the land and design the project according to that.

* Impact identification
Identify the positive and negative impacts that occur while the building of project.

- Reduce the effect of using chems unnecessarily.
- Avoid causing air pollution during construction which can cause respiratory problems

* Public participation

Communicate with the public and understand their concerns. Ask them what are the issues they face and take measures according to that.

* EIA report

Prepare an EIA report which contains the details of the project

* Compliance with laws

The project must be done by forming compliance with environmental laws and rules.

* Continous monitoring

The project should be monitored continously until the project is completed. Each changes should be recorded and updated. Required safety measures should be taken for workers as well as for the environment.

Positive impacts can include traffic reduction, reduce of air pollution, less noise pollution, easy transportation etc

Negative impact can include loss of plantation due to cutting down trees and plants, loss of habitat to human being as well as other living organisms, loss of land etc

* Impact prediction

Identify the severity and extend of the impact so that the project can be build by considering the impacts and take precautions or redesign the project

* Impact evaluation

Identify the impact and evaluate the impacts and check the how much useful and less harm it is

* Mitigation measures

Take the corresponding mitigation measures to reduce the negative impact and design build the project

It includes :-

- planting trees on both sides to not to disturb the ecosystem for cutting trees
- Rehabilate people to safer places

7a] * EIA helps to identify the impact of the environment
* It helps to provide sustainable development
* Promote the compliance with environmental laws and rules.
* Protect the plants and animals.
* Protect the biodiversity and ecosystem.
* Helps to identify impact and take precautions.
* Helps to take correct mitigation measures.
* Helps to identify the impact and redesign the project
* Helps to make better plan or decision on a project.